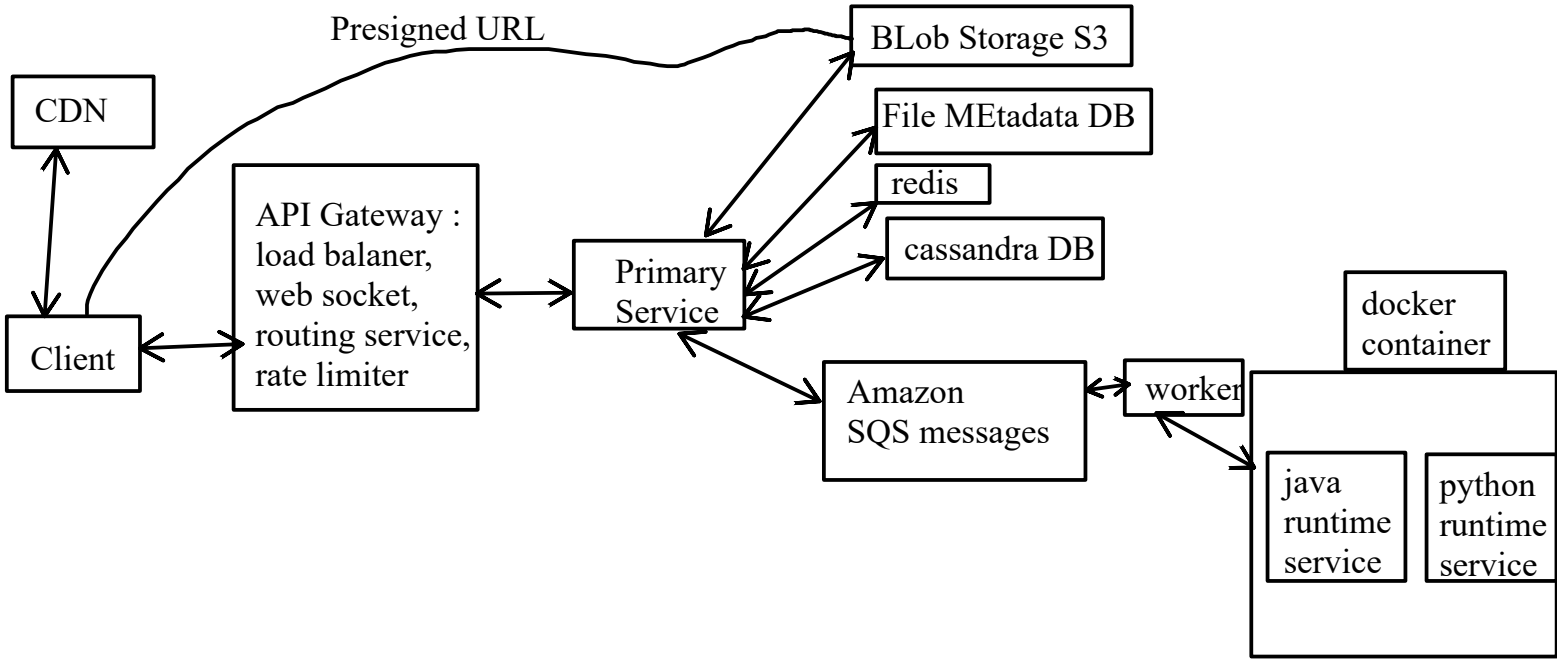
Leetcode:

Functional requirements:
user should be able to solve any problem based on his preference,
user should be able to submit soliution based on his preference,
user should be able to participate in any coding contest based on his prefernce,

Non-Functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,.
high throughput,
CAP theorem (here we need Strong Consistency for rank predictions during coding contests &
for search scenario we need to prioritise 99.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
problem,
solution,
contest

API :
POST:/problem/submit/solution
{
userID,
{filemetadata}
}
GET:/search/{problemID}

HLD :